

Last name _____

First name _____

LARSON—MATH 756—HOMEWORK WORKSHEET 07

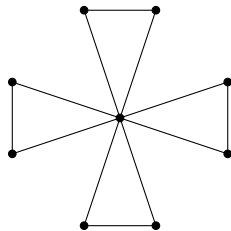


Figure 1: A flower F_4 with four petals.

Vertex Packing LPs with Added Constraints

1. Write the Vertex Packing IP model for the graph F_4 .
2. Find an optimal feasible solution for the VPIP, and argue that it is optimal.
3. Write the Vertex Packing LP model for the graph F_4 .
4. Find an optimal feasible solution for the VPLP, and argue that it is optimal.
5. Add *clique constraints* to your VPLP: add one additional constraint for each of the four triangles.
6. Find an optimal feasible solution for the updated VPLP, and argue that it is optimal.

Orthonormal Representations and Lovász Theta

7. Find any orthonormal representation $\{\hat{x}_i\}$ for F_4 . Explain.
8. Find three different unit vectors $\hat{c}$ and calculate $\max_i \frac{1}{(\hat{c}^T \hat{x}_i)^2}$ for each $\hat{c}$.
9. Find a unit vector $\vec{c}$ and orthonormal representation $\{\hat{x}_i\}$ that minimizes $\max_i \frac{1}{(\vec{c}^T \hat{x}_i)^2}$ for the graph F_4 .